

Assignment 5 - Reflection

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Through this exercise, I developed a practical understanding of Infrastructure as Code (IaC) and how Terraform is used to manage cloud infrastructure in a structured way. This exercise taught me how to use Terraform to define and manage cloud infrastructure declaratively, which improved my understanding of resource relationships. Using modules helped me organize the infrastructure logically across networking, IAM, storage, and compute components.

Compared to manually creating resources through the GCP Console, Terraform offers clear advantages such as automation, consistency, and centralized state management. While manual configuration can feel faster for small tasks, it does not scale well and is prone to errors. Terraform requires more upfront planning, especially around variables and dependencies, but this effort pays off for larger or production-like environments.

The most challenging part of this exercise was debugging issues related to Cloud Functions Gen 2, Firestore initialization, and Terraform's remote state backend. Resolving problems caused by an incorrect Firestore database setup and backend conflicts required careful analysis of error messages and Cloud Run logs. I was able to fix these issues by systematically reviewing logs and reconfiguring the Terraform backend when needed.

The Terraform features I found most valuable were modules, dependency management, and remote state. In the future, I would like to explore integrating Terraform with CI/CD pipelines and advanced state management for collaborative projects.